

11. [27th March]

(i)

10. [26th March]

(i)

9. [25th March]

- (i) Extend CYK's algorithm so that a derivation of $w \in L$ can be output in $O(|w|^3)$ time.
- (ii) Given two positive integers x and y with $x > y$, devise a DTM that computes $x \bmod y$. The time complexity of your DTM must be $O(\text{poly}(\lg_2 x))$.
- (iii) Assuming a directed graph $G(V, E)$ is given in adjacency matrix form, devise a naive polynomial time DTM to decide whether there is a path from s to t in G for $s, t \in V$.

8. [20th March]

- (i) Prove the following: Every $t(n)$ time single-tape two-dimensional DTM M_1 has an equivalent single-tape one-dimensional semi-infinite DTM M_2 whose time complexity is $O(\text{poly}(t(n)))$. (You may refer to [HU] pg 165.)
- (ii) Prove the correctness of DTM M_2 given on pg 280 of [Sip] for deciding $\{0^i 1^i \mid \text{integer } i \geq 0\}$.
- (iii) Exer 7.6 on pg 322 of [Sip].

7. [18th March]

- (i) Given a PCP instance $A : \frac{10111}{10}, \frac{1}{111}, \frac{10}{0}$, construct a CFG G' such that $f(\langle A \rangle) = \langle G' \rangle$, where f is a recursive function that reduces a PCP instance to an instance of $L' = \{\langle G \rangle \mid \text{CFG } G \text{ is ambiguous}\}$. Prove $\langle G' \rangle \in L'$.
- (ii) Given a PCP instance $A : \frac{10}{101}, \frac{011}{11}, \frac{101}{011}$, construct a CFG G' such that $f(\langle A \rangle) = \langle G' \rangle$, where f is a recursive function that reduces a PCP instance to an instance of $L' = \{\langle G \rangle \mid \text{CFG } G \text{ is ambiguous}\}$. Prove $\langle G' \rangle \notin L'$.

6. [13th March]

- (i) Define all dominos in MPCP instance A that corresponds to $\langle M, w \rangle$, where M is a TM as defined in [12th March] problem and $w = 110011$, as in the reduction from L_u to MPCP problem. Provide an accepting computation history H of M on w . By placing dominos of A according to H , show there is a solution to A .
- (ii) Prove at least one domino from groups III and IV listed on pg 197 of [HU] exists in any solution for an MPCP instance constructed from an instance of L_u .
- (iii) In reducing MPCP problem to PCP problem, determine whether the number of α 's introduced in constructing a PCP instance B can be reduced. State the significance of positioning each of the remaining α 's.

Let B be the PCP instance corresponding to MPCP instance $A : \frac{10111}{10}, \frac{1}{111}, \frac{10}{0}$ in the MPCP to PCP reduction. Find a solution sol_B for the PCP instance B . Based on sol_B , construct a solution for MPCP.

5. [12th March]

Let M be a TM with $L(M) = \{w \mid w \in \{0, 1\}^* \text{ both starts and ends with } 11\}$.

- (i) Following the proof of [HU] Lemma 8.6, give two PDAs P_1 and P_2 such that $L(P_1) \cap L(P_2)$ is the set comprising all the valid computation histories of M .
- (ii) Following the proof of [HU] Lemma 8.7, give a PDA P such that $L(P)$ is the set comprising all the invalid computation histories of M .

4. [11th March]

- (i) In the context of Rice's theorems, define two examples for each of the following properties: (i) non-trivial, (ii) monotone, and (iii) non-monotone. (These example properties must not have been mentioned in lectures.) And, prove the same.
- (ii) For any one of the languages, say L'' , that was shown to be undecidable in previous lectures, by designing a relevant non-trivial property Γ and by applying Rice's first theorem, argue L'' is undecidable.
- (iii) For any one of the languages, say L'' , that was shown to be not Turing-recognizable in previous lectures, by designing a relevant non-monotone property Γ and by applying Rice's second theorem, argue L'' is not Turing-recognizable.

3. [6th March]

- (i) Give a mapping reduction from $\overline{L}_u$ to $L_{decid} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is decidable}\}$. What can you conclude from this reduction?
- (ii) Show $\overline{L}_u \leq_m \overline{L}_{equal}$ without using $L_u \leq_m L_{equal}$.
- (iii) Given a binary encoding $\langle M \rangle$ of a TM M and a string $w \in \Sigma^*$ on the input tape (with $\langle M \rangle$ and w separated with a special symbol) of a TM M' , describe all the details of M' which "on input w' accepts w' if M accepts w ".

Give a TM that computes a recursive function f such that $\forall_{\langle M, w \rangle} f(\langle M, w \rangle) = \langle M' \rangle$, where M, w , and M' are as defined above.

Construct a UTM M'' that runs $\langle M' \rangle$ from the above problem on input $w'' \in \Sigma^*$ and prints on its output tape " M' accepts w'' " if M' indeed accepts w'' .

2. [5th March]

- (i) Problem 5.7 on page 239 from [Sip].
- (ii) Given a string $w \in \{0, 1\}^*$ on the input tape of a TM M'' , implement M'' so that M'' verifies whether w represents the binary encoding of a TM.
- (iii) Given a valid binary encoding $\langle M \rangle$ of a TM M on the input tape of a TM M'' , assuming M has only one reject state, implement M'' so that it outputs a binary encoding $\langle M' \rangle$ of a TM M' wherein M' is same as M except that upon entering reject state M' loops.

1. [4th March]

- (i) Prove that Kleen star is closed for the set of Turing-recognizable languages.
- (ii) Given two TMs and $w \in \Sigma^*$, write the description of a Turing machine that dovetails the computations of these machines on input w .